Experiment No. 5:

Implement Hidden Markov Model (HMM) for Real-World Problem (Weather Forecasting)

1. Aim:

To understand and implement the **Hidden Markov Model (HMM)** for **weather forecasting**.

2. Objectives:

1. To understand the working principle of **Hidden Markov Model (HMM)**.
2. To perform **probabilistic inference** using the **Forward Algorithm**.
3. To use **HMM** for **weather forecasting** based on observable data.
4. To predict future hidden states based on observation sequences.

3. Course Outcome:

CO4

Implement HMM for real world problem (example weatherforcasting)

4. Problem Statement:

You are given a real-world problem where:

- There are **two possible hidden states (Weather)**:
  - **Rainy**
  - **Sunny**
- You cannot directly observe the weather, but you can observe the **activities** of a person:
  - **Walk**
  - **Shop**
  - **Clean**

The task is to implement a **Hidden Markov Model (HMM)** to predict the weather based on the sequence of activities.

5. Hidden Markov Model (HMM) Details:

Components of HMM:

A **Hidden Markov Model (HMM)** consists of:

Component	Meaning
Hidden States (H)	States that are not directly observable but can be inferred. In our case: **Rainy** or **Sunny**.
Observable States (O)	The output that we can observe. In our case: **Walk, Shop, Clean**.
Transition Probabilities (A)	Probability of transitioning from one hidden state to another. (Rainy → Sunny, Sunny → Rainy, etc.)
Emission Probabilities (B)	Probability of observing a particular activity given a hidden state. (e.g., P(Walk / Sunny))
Initial Probabilities (π)	Probability of starting in a particular state (e.g., 60% Rainy, 40% Sunny).

6. Parameters of HMM:

Here are the parameters we use:

1. States (Hidden States):

- $S = \{\text{Rainy,Sunny}\}$

2. Observations (Activities):

- $O = \{\text{Walk,Shop,Clean}\}$

3. Initial Probabilities (π):

- $P(Rainy) = 0.6$
- $P(Sunny) = 0.4$

4. Transition Probabilities (A):

From/To	Rainy	Sunny
Rainy	0.7	0.3
Sunny	0.4	0.6

5. Emission Probabilities (B):

Activity/Weather	Rainy	Sunny
Walk	0.1	0.6
Shop	0.4	0.3
Clean	0.5	0.1

7. Forward Algorithm (For Prediction)

The **Forward Algorithm** is used to:

- **Calculate the probability of an observed sequence** given the model.
- Predict the hidden states (Weather) based on observed activities.

Mathematical Formula for Forward Algorithm:
The probability of observing a sequence O given the model λ is:

$$P(O|\lambda) = \sum_{i=1}^{N} \alpha_T(i)$$

Where:

- $\alpha_t(i)$ = Probability of observing the sequence until time t and being in state i.
- T = Length of the sequence.
- N = Number of hidden states.

8. Python Code implementation

Python code for **HMM Weather Prediction**.

9. Output

The output will be as:

```
The probability of observing the sequence ['Walk', 'Shop',
'Clean'] is: 0.03186
```

This means the probability of observing the sequence **(Walk → Shop → Clean)** based on the hidden weather conditions is **3.18%**.

10. Result Analysis

The **Hidden Markov Model (HMM)** successfully predicted the hidden weather based on observable activities:

Activity	Most Likely Weather
Walk	Sunny
Shop	Rainy
Clean	Rainy

11. Viva Questions

Question	Answer
1. What is Hidden Markov Model (HMM)?	HMM is a statistical model used to describe a process with hidden states based on observable outputs.
2. What are the components of HMM?	Hidden States, Observable States, Transition Probability, Emission Probability, Initial Probability.
3. Why is it called 'Hidden' Markov Model?	Because the actual states (like weather) are hidden and can only be inferred using observations.
4. What is the Forward Algorithm used for?	To calculate the probability of an observed sequence given a model.
5. Can HMM be used in Speech Recognition?	Yes, HMM is widely used in Speech Recognition, Gesture Recognition, and POS Tagging.

12. Conclusion

- Implemented the **Hidden Markov Model (HMM)** for **weather forecasting** using Python.
- The **Forward Algorithm** was used to calculate the probability of a sequence of observable states.
- **HMMs** are powerful for problems where **the internal process is hidden** but output can be observed.

13. Reference Books:

1. Kevin Murphy, *Machine Learning - A Probabilistic Perspective*, MIT Press, 2012.
2. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
3. Tom Mitchell, *Machine Learning*, McGraw Hill, 1997.